Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 12: Reproduction in Plants

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	12 — Reproduction in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This is the hardest tier: every question demands multi-step synthesis across asexual modes (budding, fragmentation, spores, vegetative propagation) and the full sexual pathway (flower parts, pollination, fertilisation, fruit/seed formation, dispersal). Distractors are built from the deepest confusions — pollen-transfer versus gamete-fusion, ovule->seed versus ovary->fruit versus zygote->embryo, clone-uniformity versus seed-variation, spore (no fusion) versus seed (post-fusion), agent-matched adaptations, and the strict order of life-cycle events under experimental manipulation — so eliminate by reasoning, not by guessing. Do not write on this paper — mark answers on your answer sheet.

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1. A breeder keeps a prized apple variety alive in two ways at once: a row grafted onto wild rootstocks, and a row grown each year from that variety's own seeds. After ten years she finds the grafted row still gives the exact same famous fruit, while the seed row has drifted into many slightly different fruits. The single principle that explains BOTH outcomes together is that:

(A) the wild rootstock slowly passed its fruit traits up into the grafted shoot, while the seeds stayed pure
(B) grafted shoots reproduce by spores that fix the traits, while seeds reproduce by buds that change them
(C) grafting copies one parent's traits unchanged (a clone), while seeds come from fertilisation that reshuffles traits, so only the seed row can vary
(D) the seed row was cross-pollinated and so stayed identical, while grafting forced new variation each year

2. A student is shown four events and told that only ONE of them ends with two cells joining into a single cell. The events are: a yeast bud pinching off, a Spirogyra filament snapping in two, a fern spore landing and growing, and a pollen tube delivering a male gamete to an egg. The event that involves cells joining is:

(A) the yeast bud pinching off, because the bud fuses back with the parent before leaving

(B) the Spirogyra filament snapping, because the two pieces rejoin to make one body

(C) the fern spore growing, because two spores must fuse before a fern can form

(D) the pollen tube delivering the male gamete to the egg, because that fusion is fertilisation

3. A gardener plants two identical-looking potato pieces. Piece 1 has its single eye intact; piece 2 has the eye carefully scooped out but keeps all its surrounding flesh. Both are buried in the same moist tray. Three weeks later only piece 1 has a shoot. A classmate concludes the difference proves the eye supplies the stored food the shoot needs. The classmate's conclusion is:

(A) right, because the eye holds the tuber's food and the scooped flesh is empty

(B) wrong, because the eye is a bud that grows the shoot; the flesh, still present in piece 2, is the food store

(C) right, because the eye contains the ovule that becomes the new shoot

(D) wrong, because piece 2 should still sprout from its flesh given more time

4. A flower has a normal stigma, style and ovary, healthy stamens, and is pollinated perfectly by bees. Yet a biologist predicts with confidence that this flower can NEVER form a seed. The single observation that would justify her prediction is that:

(A) the anthers make sticky pollen, so the bees cannot carry it far enough

(B) the style is too long, so the pollen tube can never reach the ovary

(C) the flower is bisexual, and bisexual flowers cannot set seed

(D) the ovary contains no ovules, so even after fertilisation there is nothing to become a seed

5. Two ponds are sampled. In pond A the green alga Spirogyra is multiplying so fast the water turns green; in pond B a thick-walled resistant body is found that only grows when the dry season ends. A student says both ponds show the same mode of reproduction. The sharpest correction is that:

(A) both show budding, the alga budding fast and the resistant body being a slow bud

(B) pond A shows fragmentation (the body breaks into growing pieces) while pond B shows a spore, which waits protected and grows only when conditions improve

(C) pond A shows spore formation and pond B shows fragmentation, the reverse of what is claimed

(D) both show seed formation, since each new body grows from a stored unit

6. A horticulturist must mass-produce a new banana that bears NO usable seeds, and wants every plant identical to the original. A junior suggests collecting the tiny specks inside the pulp and sowing them. The senior rejects this and instead splits off shoots from the underground stem. The senior's choice is best because:

(A) the seedless banana cannot be raised from seed, and splitting the underground stem is vegetative propagation, giving identical clones

(B) the specks are spores that would grow into a different fungus, so the stem is safer

(C) sowing the specks would give clones too, but the underground stem grows faster

(D) the underground stem reproduces sexually, which keeps the new plants seedless

7. Before a bisexual flower opens, a researcher does three things to it: removes all the anthers, leaves it uncovered beside many plants of the same kind, and waits. It later sets seed with viable embryos. The MINIMUM set of processes that must have occurred is:

(A) self-pollination, then fertilisation, then seed formation

(B) cross-pollination, then fertilisation, then seed formation

(C) cross-pollination alone, since seeds can form straight from pollination

(D) fertilisation, then cross-pollination, then seed formation

8. A flower's anthers shed all their pollen on Monday, but its own stigma does not become sticky and receptive until Friday. Over those days bees visit constantly. The most likely effect of this timing on the flower's seeds is that they will be:

(A) formed only from its own pollen, because bisexual flowers always self-pollinate first

(B) unable to form at all, because the stigma missed the pollen entirely

(C) formed mostly from another flower's pollen, because the flower's own pollen was gone before its stigma was ready

(D) formed from spores instead, because the timing mismatch blocks pollination

9. A child traces a single male gamete from the moment its pollen grain lands until it meets the egg, listing the structures it passes through. The correct ordered list of structures is:

(A) anther, then filament, then ovary, then ovule to the egg

(B) stigma, then ovary, then style, then back up to the ovule

(C) stigma, then down through the style, into the ovary, and into the ovule to the egg

(D) filament, then style, then stigma, then ovule to the egg

10. A botanist cuts open one fruit and finds exactly five seeds, each holding a tiny embryo. From this single fruit she draws the most direct conclusion about the parent flower. That conclusion is that the ovary held:

(A) exactly five anthers, one ripening into each seed

(B) one ovule that split into five seeds after fertilisation

(C) five zygotes that formed before pollination even began

(D) at least five ovules, each of which was fertilised and became a seed

11. Two adjacent fields of one crop are wiped out by a brand-new disease in different patterns. Field 1 (all grown from cuttings of a single elite plant) dies almost completely and all at once; field 2 (grown from that plant's seeds) loses only about half its plants. The deepest single reason for the contrast is that:

(A) the seed plants were sprayed and the cutting plants were not

(B) the cutting field is one set of identical clones sharing one weakness, while the seed field carries varied traits so some plants resisted

(C) cuttings reproduce by spores that attract disease, while seeds repel it

(D) the cutting plants were older, and older plants always die first

12. A maize plant carries a feathery tassel at its top that releases pollen, and a separate cob lower down whose silky threads catch that pollen. A student claims this is one huge bisexual flower spread over the plant. The accurate description is that maize has:

(A) one bisexual flower whose male and female parts are simply far apart

(B) no real flowers, since the pollen comes from leaves

(C) separate male flowers (in the tassel) and female flowers (on the cob), both on the same plant

(D) only male flowers, the cob being a fruit that needs no female flower

13. A gardener buries part of a low, still-attached jasmine branch in soil; roots grow there, and only afterwards does she cut the rooted portion free as a new plant. A neighbour insists this is fragmentation because a piece was finally cut off. The neighbour is wrong because:

(A) this is actually spore formation, since the buried stem makes spores in the dark

(B) this is cross-pollination between the branch and the soil minerals

(C) the branch rooted while still joined to the parent (layering, a vegetative method); fragmentation means a body first breaks into pieces that then grow

(D) the neighbour is right, because any method that ends in a cut is fragmentation

14. A science fair has three exhibits side by side: a damp moss cushion shedding fine units from tiny capsules, a Bryophyllum plant whose leaf edges are dotted with miniature plantlets, and a tray of mustard plants grown from sown seeds. A judge asks which two exhibits use the SAME basic kind of reproduction. The correct pairing and reason is:

(A) the Bryophyllum and the mustard, because both grow their new plants from buds

(B) the moss and the mustard, because both scatter tiny units that are really seeds

(C) the moss and the Bryophyllum, because both are asexual (spores from capsules, buds on leaf edges) with no fusion of gametes, while the mustard from seeds is sexual

(D) all three, because any way of making a new plant counts as the same kind of reproduction

15. A diagram of one flower labels four parts: W makes the pollen, X is a slender stalk holding W up, Y is a sticky tip where pollen lands, and Z houses the ovules. Naming them in the order W, X, Y, Z gives:

(A) filament, anther, ovary, stigma
(B) stigma, style, anther, ovary
(C) ovary, filament, stigma, anther
(D) anther, filament, stigma, ovary

16. A flower is bagged before it opens so no insect or breeze can reach it, yet it still sets seed. A second, identical flower has its anthers removed before bagging and sets NO seed at all. Comparing the two results, the strongest conclusion is that:

(A) the first flower was cross-pollinated, because the bag let a neighbour's pollen slip in
(B) both flowers formed seeds without any pollination, the second simply faster
(C) the first flower self-pollinated using its own pollen, since removing the anthers (its only pollen source) in the second flower stopped seeds entirely
(D) the first flower made seeds from spores released inside the bag

17. Following one ovule through the whole sexual cycle of a flowering plant, a student must say what it ends up as and what is inside it. The correct end state is that the ovule becomes a:

(A) fruit, inside which all the other ovules are enclosed
(B) fresh stigma, ready to catch pollen for the next flower
(C) pollen grain, to be carried away in the next round of pollination
(D) seed, inside which the zygote it held has grown into an embryo

18. A student groups four organisms by how a new individual arises: a Hydra growing a bulge that detaches, yeast pinching off an outgrowth, a moss scattering tiny units from a capsule, and a Spirogyra filament breaking into pieces. The student must pick the TWO that share the very same process. They are:

(A) moss and Spirogyra, because both reproduce by budding
(B) Hydra and yeast, because both reproduce by budding (an outgrowth forms and separates)
(C) Hydra and Spirogyra, because both break the parent body into pieces
(D) yeast and moss, because both release spores from a case

19. A grower lists three things she wants: (i) a winter batch of money plants all identical to one prized vine, (ii) varied tomato plants for a breeding trial, and (iii) more ferns for a shady corner. The methods best matched to (i), (ii), (iii) are, in order:

(A) sowing seeds, stem cuttings, scattering spores
(B) scattering spores, sowing seeds, stem cuttings
(C) stem cuttings, sowing seeds, scattering spores
(D) stem cuttings, scattering spores, sowing seeds

20. A flower with bright petals, sweet scent, deep nectar and sticky pollen grows beside a submerged water plant that releases light pollen straight into the moving current. A student says both must use the same pollinating agent. The correct matching of agents to the two plants is:

(A) wind for the showy flower and insects for the submerged plant

(B) water for both, since both release pollen into the open

(C) insects for the showy nectar-rich flower and water for the submerged plant

(D) insects for both, the submerged plant attracting underwater bees

21. After a pea flower is fertilised, the petals and stamens wither and drop while the flower's base swells and lengthens into a pod holding several peas. Identifying the swelling pod and what each pea grew from, the correct statement is that:

(A) the stigma became the pod, and each pea grew from an anther

(B) the filament became the pod, and each pea grew from a pollen grain

(C) the anther became the pod, and each pea grew from a zygote alone with no ovule

(D) the ovary became the pod (fruit), and each pea (seed) grew from an ovule

22. A student writes a rule: 'Any flower that has both stamens and a pistil is certain to self-pollinate.' A teacher gives one real flower that disproves the rule. The disproving flower is one that:

(A) has both parts and immediately drops pollen onto its own stigma every time

(B) has both parts, yet sheds all its pollen before its own stigma becomes receptive, so its pollen cannot reach its own stigma

(C) has only stamens and no pistil, so it cannot self-pollinate

(D) reproduces by spores instead of pollen, so the rule does not apply

23. A seed bears a tuft of fine hairs like a parachute, a second fruit has a thick spongy buoyant husk, and a third fruit is studded with tiny stiff hooks. A student says all three are adapted to one dispersal agent. The correct matching, in order, is:

(A) wind, water, animals

(B) water, wind, animals

(C) animals, wind, water

(D) wind, animals, water

24. A flower is dusted heavily with pollen by bees, the pollen tubes are seen growing down the style, yet weeks later not one fruit forms. Of all explanations a student offers, the single one that fits is that:

(A) pollination failed, since pollen plainly never reached the stigma

(B) fertilisation failed — the male gamete never fused with an egg, so no fruit developed

(C) the ovary turned straight into seeds, skipping the fruit

(D) seed dispersal happened before the fruit could form

25. Four numbered events of a flowering plant's life are: (1) the ovary ripens into a fruit holding seeds; (2) a male gamete fuses with an egg; (3) pollen reaches a stigma; (4) a seed germinates into a new plant. The correct order is:

(A) 2, then 3, then 1, then 4

(B) 3, then 1, then 2, then 4

(C) 3, then 2, then 1, then 4

(D) 1, then 3, then 2, then 4

26. A wind-pollinated grass and an insect-pollinated apple tree grow side by side. The grass releases enormous clouds of light, dry pollen; the apple makes far less, stickier pollen. The deepest reason the grass needs so much more pollen is that:

(A) wind pollen is heavier, so more grains are needed to lift it into the air

(B) insects eat most of the grass pollen before it can reach a stigma

(C) wind scatters pollen at random, so a huge surplus is needed for any grain to chance upon a stigma

(D) the grass wastes pollen on purpose to attract more pollinators

27. A student argues: 'A fern on a damp wall and a potato sprouting in a dark cupboard reproduce in exactly the same way, because neither uses a flower.' The sharpest correction is that:

(A) they are the same, since any plant without a flower must use spores

(B) the fern uses buds and the potato uses spores, the reverse of the truth

(C) lacking a flower is shared, but the fern uses spores while the potato uses buds in its tuber (vegetative propagation) — different asexual routes

(D) the fern reproduces sexually by seeds and the potato by fragmentation

28. A grower wants to be certain a particular tomato flower's seeds come only from its own pollen, with zero outside pollen. The single experimental step that guarantees this is to:

(A) remove the flower's anthers and leave it open near other tomato plants

(B) bag the unopened flower so no insect or breeze can bring in any outside pollen

(C) place the open flower in a breezy spot full of visiting bees

(D) scatter spores from the same plant onto the flower's stigma

29. A student is given five cards to sort into 'asexual' and 'sexual': a strawberry runner that roots and forms a new plant, a sugarcane stem cutting, a bread-mould spore, an apple grown from its seed, and a sweet-potato root sprout. After sorting, the student claims the apple-from-seed card is the only one that belongs in 'sexual'. This claim is:

(A) wrong, because the spore card also belongs in sexual, since spores form after gametes fuse

(B) correct, because only the apple from seed involves fertilisation, while the runner, cutting, spore and root sprout are all asexual

(C) wrong, because the strawberry runner belongs in sexual, since runners come from flowers

(D) correct, but only because apples happen to be the one plant that can make seeds

30. Immediately after a male gamete fuses with the egg in an ovule, a student is asked what the very first new structure is, and what it later becomes. The correct pair is that:

(A) the seed coat forms first, and it later becomes the fruit

(B) the fruit forms first, and it later becomes the seed

(C) the zygote forms first, and it later grows into the embryo inside the seed

(D) a new pollen grain forms first, and it later becomes the embryo

31. A nursery owner grafts a shoot of a sweet apple variety onto a hardy rootstock and advertises that the tree will now bear a brand-new apple that blends both plants' fruit flavours. His claim is:

(A) wrong, because the grafted shoot keeps the chosen variety's own fruit traits rather than blending with the rootstock

(B) right, because grafting fuses the gametes of the two plants into a new variety

(C) right, because the rootstock's pollen reaches the shoot's flowers and mixes the traits

(D) wrong, because grafting makes the shoot produce spores instead of fruit

32. A plant always drops its seeds straight down at its own base, with no hooks, hairs or buoyant husk to carry them. Over many years, the clearest disadvantage compared with a plant whose seeds travel is that:

(A) its crowded seedlings compete with the parent and one another for light, water, minerals and space

(B) its seedlings cannot photosynthesise while close to the parent

(C) its seeds turn back into spores when they land near the parent

(D) its seedlings get too much room and starve from being spread out

33. A student must pick the ONE observation that, by itself, proves a particular new plant arose by sexual reproduction and not by any vegetative route. The proving observation is that the new plant:

(A) sprouted from a bud on the margin of a leaf laid on soil

(B) grew from a seed that had formed inside a fruit after a flower was fertilised

(C) rooted from a stem cutting kept in a glass of water

(D) grew up from a buried piece of underground stem

34. A flower has its entire ovary carefully removed before it opens, but everything else is left intact, and it is then pollinated normally by bees. A student lists what can and cannot still happen. The correct statement is that:

(A) pollen can no longer be made, because the anthers sit inside the ovary

(B) the stigma can no longer receive pollen, because it grows out of the ovary

(C) seeds can still form, because the stigma alone can turn pollen into seeds

(D) pollen can still land on the stigma, but no seed can ever form, because the ovules that become seeds were inside the removed ovary

35. A spore from a fern and a pea seed are placed side by side. A student claims they are basically the same kind of reproductive unit. The single most fundamental difference is that:

(A) the seed formed after fertilisation (gametes fused), while the spore is an asexual unit formed with no gamete fusion

(B) the spore formed after fertilisation, while the seed needs no gametes at all

(C) both formed after pollination, the only difference being their size

(D) the seed is always lighter and more wind-blown than the spore

36. Two flowers on different plants are described. Flower P is large, brightly coloured, fragrant and offers nectar; Flower Q is tiny, dull-green, scentless and dangles loosely, shedding clouds of light dry pollen. Predicting their agents and one extra adaptation each, the best matched pair is:

(A) P is wind-pollinated (its scent draws the breeze), Q is insect-pollinated (its dull colour attracts bees)

(B) P is insect-pollinated (sticky pollen rewards the visitor), Q is wind-pollinated (huge amounts of light pollen for random scattering)

(C) both are water-pollinated, P floating and Q sinking

(D) both are insect-pollinated, differing only in how big the insects are

37. A student must trace the female parts of a flower from the surface where pollen lands down to where future seeds form, naming them in order. The correct order is:

(A) stigma, then ovary, then style

(B) stigma, then style, then ovary (with ovules inside)

(C) anther, then filament, then ovary (D) ovary, then style, then stigma

38. A field of one crop is raised entirely from cuttings of a single elite plant. A breeder predicts that, no matter how many seasons pass, this method alone will never throw up a plant with a brand-new useful trait the elite parent lacked. The breeder's prediction is:

(A) sound, because cuttings copy one parent exactly, with no fertilisation to reshuffle traits, so no genuinely new trait can appear

(B) unsound, because repeated cuttings slowly accumulate new traits over the seasons

(C) unsound, because each cutting secretly mixes in pollen from a neighbouring plant

(D) sound, but only because this particular crop happens to be seedless

39. A botanist compares two habits in one species: some plants regularly self-pollinate, others are cross-pollinated by insects. She notes one clear benefit each habit gives. The benefit correctly matched to each habit is that:

(A) self-pollination gives the most varied offspring, while cross-pollination guarantees seed set without any pollinator

(B) self-pollination produces spores cheaply, while cross-pollination produces seeds slowly

(C) both habits give exactly the same offspring, so neither has any special benefit

(D) self-pollination lets a plant set seed even with no pollinator or partner, while cross-pollination brings together two parents' traits, giving more varied offspring

40. A botanist insists pollination must come strictly before fertilisation and never the other way round. The clearest reason is that:

(A) the male gamete is delivered by pollen, so pollen must first reach the stigma before a gamete can fuse with the egg

(B) the ovary must ripen into a fruit before the gametes are allowed to fuse

(C) the petals must fall off before pollen is permitted to land on the stigma

(D) seed dispersal must happen first so the gametes have room to meet

41. Inside one fruit a student finds many seeds; in a second fruit she finds just one large seed (a stone). A classmate concludes the second flower must have used a completely different reproductive process. The best correction is that:

(A) the one-seeded fruit formed by vegetative propagation, the many-seeded one by seeds

(B) both used the same sexual process; the many-seeded fruit's ovary simply held many ovules, the one-seeded fruit's ovary held a single ovule

(C) the one-seeded fruit grew from a spore, while the many-seeded one grew from a flower

(D) the many-seeded fruit was self-pollinated and the one-seeded fruit reproduced asexually

42. A hiker returns from a walk with small hooked fruits stuck to her socks; that evening a bird in the orchard swallows sweet berries and later voids the hard seeds far off; meanwhile a ripe pod in the garden dries and snaps, flinging its seeds about. The dispersal agents for these three cases are, in order:

(A) animals (hooks on clothing), animals (eaten and passed), and explosion (the bursting pod)

(B) wind, animals, and water

(C) animals, wind, and explosion

(D) explosion, animals, and wind

43. A flower's anthers are removed before it opens; it is bagged so no outside pollen can enter; yet weeks later it still appears to set fruit. A student calls this a clear case of self-pollination. The sharpest correction is that:

(A) it is self-pollination, because the bag trapped the flower's own pollen inside

(B) it is cross-pollination, because the bag let a little neighbour pollen slip through

(C) it is spore formation, triggered by the dark conditions inside the bag

(D) it cannot be self-pollination, since the flower had no pollen of its own and none could enter — so any seed-bearing fruit here would be impossible, signalling an error in the observation

44. A student pairs each organism with its asexual mode and gets one pair wrong. The pairs are: Rhizopus (bread mould) with spore formation; Spirogyra with fragmentation; yeast with budding; potato with spore formation. The wrong pair is:

(A) Rhizopus with spore formation, because bread mould actually buds

(B) Spirogyra with fragmentation, because the alga actually makes spores

(C) yeast with budding, because yeast actually reproduces by fragmentation

(D) potato with spore formation, because a potato sprouts from buds in its tuber (vegetative propagation), not from spores

45. A flower is described as 'unisexual male'. A student must say which single later event is impossible for this flower acting alone, and why. The correct statement is that:

(A) it cannot form seeds by itself, because it has no pistil (no ovary or ovules) to be fertilised

(B) it cannot release pollen, because male flowers have no anthers

(C) it cannot be pollinated, because its pollen has nowhere to go

(D) it cannot reproduce at all, because male flowers are always sterile

46. A nursery raises four crops by these exact parts: a dahlia from a swollen root, an onion from a bulb (a shortened underground stem), a rose from a leaf-and-bud cutting, and a sunflower from its dry seed. A trainee must point to the crop whose part is NOT a route of vegetative propagation, and say why. The correct answer is:

(A) the sunflower's seed, because a seed is the product of fertilisation (sexual reproduction), unlike the vegetative root, bulb and cutting

(B) the dahlia's root, because roots can never give rise to a new plant

(C) the onion's bulb, because a bulb is a fruit and not a stem at all

(D) the rose's cutting, because cuttings always grow from hidden seeds inside the stem

47. After fertilisation a student must match three starting structures to their fates: the ovule, the ovary, and the zygote. The correct matching is:

(A) ovule becomes the fruit, ovary becomes the seed, zygote becomes the embryo

(B) ovule becomes the embryo, ovary becomes the fruit, zygote becomes the seed

(C) ovule becomes the seed, ovary becomes the fruit, zygote becomes the embryo

(D) ovule becomes the seed, ovary becomes the embryo, zygote becomes the fruit

48. A grower takes many cuttings from one money-plant vine and raises them in different spots: some in bright light, some in shade, some in rich soil, some in poor soil. The shaded ones grow smaller and paler. A student says this proves the cuttings are now genetically different from one another. The student is:

(A) right, because growing in different light permanently changes each cutting's traits

(B) right, because the poor-soil cuttings switched to sexual reproduction to survive

(C) wrong, because the cuttings actually grew from seeds that varied from the start

(D) wrong, because all the cuttings are clones of the one parent; the size and colour differences come from the environment, not from different traits

49. An aquatic plant grows fully submerged and releases its pollen directly into the moving water; a grass on the bank dangles loose anthers that shake pollen into the breeze. The pollinating agents for the submerged plant and the grass are, respectively:

(A) wind for the submerged plant and water for the grass

(B) insects for both, attracted by the open anthers

(C) water for the submerged plant and wind for the grass

(D) water for both, since both release pollen loosely

50. To test whether pollination alone is enough to make a fruit, a student dusts pollen onto the stigmas of ten flowers, then keeps half of them in conditions that block the pollen tubes from reaching the ovules. None of that half forms fruit, while the other half do. The clearest conclusion is that:

(A) pollination alone makes the fruit, and the blocked half failed only because the pollen was old

(B) pollination is not enough on its own; fertilisation (the gamete fusion that the blocked tubes prevented) must also occur for a fruit to form

(C) the blocked flowers formed fruit without seeds, proving fruit needs no fertilisation

(D) the blocked flowers reproduced by spores instead, which is why no fruit appeared

51. A gardener wants every new plant of a prized seedless ornamental to be identical to the one parent AND wants them quickly, within a single season. The method meeting both aims is:

(A) taking and planting stem cuttings from that parent

(B) collecting and sowing the parent's seeds each season

(C) letting wind cross-pollinate it with neighbouring plants

(D) waiting for the ornamental to release spores

52. A diary records a flower over several days: Day 1 a bee leaves pollen on the stigma; Day 3 a pollen tube reaches an ovule and the male gamete fuses with the egg; Day 4 a zygote is seen; Day 20 a ripe fruit holds seeds. The student must mark the day the new plant truly began. The correct day and reason is:

(A) Day 1, because leaving pollen on the stigma is the start of the new plant

(B) Day 20, because the new plant begins only once the fruit and seeds are ready

(C) Day 4, because the new plant begins when the zygote is first seen, not when the gametes fuse

(D) Day 3, because the new plant begins when the gametes fuse in fertilisation, forming the zygote seen the next day

53. A student must choose the seed feature that would be LEAST helpful for dispersal by wind. The least helpful feature is:

(A) a tuft of fine hairs forming a parachute(B) thin, flat papery wings

(C) a very small, light body (D) a heavy seed with a thick, stony coat

54. A flower has bright petals, working stamens, and a pistil — but its ovary, though present, contains not a single ovule. Even after flawless pollination and a pollen tube reaching the ovary, this flower's clear limit is that it cannot:

(A) form any seed, because with no ovules there is nothing to develop into a seed after fertilisation

(B) release any pollen, because the empty ovary blocks the anthers

(C) have its stigma receive pollen, because no ovules means no stigma

(D) be visited by insects, because they sense the empty ovary and stay away

55. A student writes: 'Pollen lands on the filament during pollination.' A teacher asks for the precise correction. The precise correction is that:

(A) pollen lands on the stigma; the filament is only the stalk that holds the anther aloft

(B) the statement is correct, because the filament is sticky and traps pollen

(C) pollen lands on the anther, since that is also where it is stored

(D) pollen lands inside the ovary, not on the filament

56. A breeder deliberately dusts pollen from one chosen strawberry variety onto the stigma of a second chosen variety, hoping the seeds will combine the best traits of both. Naming both the pollen-transfer step and the overall deliberate practice, the correct pair is:

(A) the transfer step is cross-pollination, and the deliberate crossing of two varieties is hybridisation

(B) the transfer step is fertilisation, and the deliberate crossing is self-pollination

(C) the transfer step is self-pollination, and the deliberate crossing is fragmentation

(D) the transfer step is pollination, and the deliberate crossing is budding

57. A student must pick the ONE observation that would most convincingly show a new plant arose by VEGETATIVE propagation rather than from a seed. The convincing observation is that the new plant:

(A) grew from a unit that had formed inside a fruit after a flower was fertilised

(B) appeared after bees visited the parent's bright flowers all season

(C) started from a tiny embryo found inside a hard seed coat

(D) grew directly from a bud on a buried piece of the parent's underground stem, with no flower, fruit or seed involved

58. A juicy, brightly coloured, sweet-smelling fruit is found high in a tree, far from any water and with no hooks, hairs or buoyant husk. A student must explain why such fruits are regarded as an adaptation for animal dispersal rather than wind or water. The best explanation is that:

(A) the bright colour makes the whole fruit light enough to be blown by the wind

(B) animals are drawn to eat the tasty fruit and then drop or pass the tough seeds far from the parent

(C) the juice helps the entire fruit float away down a river

(D) the sweetness makes the fruit burst open and fling its seeds out

59. A student says budding in yeast and fragmentation in Spirogyra are 'just two names for the same thing, since in both something separates from the parent.' The sharpest correction is that:

(A) they are indeed the same, since separation occurs in both

(B) budding breaks the body into pieces while fragmentation grows one outgrowth, the reverse of the truth

(C) both are really spore formation, since a parent cell is involved in each

(D) in budding one small outgrowth forms and pinches off, while in fragmentation the body itself breaks into pieces that each grow — different processes

60. A student finally summarises the difference between two beds of one crop: Bed A grown from seeds occasionally shows a plant with a useful new trait the parent never had, while Bed B grown from cuttings never does. The single correct reason is that:

(A) seeds arise from fertilisation, which reshuffles traits and can yield variation, while cuttings only clone the one parent

(B) cuttings produce spores that erase the parent's traits each generation

(C) seeds skip fertilisation and so invent brand-new traits out of nothing

(D) cuttings need two parents while seeds need only one

Solutions — Science (Class 7), Chapter 12: Reproduction in Plants — Paper 5

Paper 5 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	D	31	A	41	B	51	A
2	D	12	C	22	B	32	A	42	A	52	D
3	B	13	C	23	A	33	B	43	D	53	D
4	D	14	C	24	B	34	D	44	D	54	A
5	B	15	D	25	C	35	A	45	A	55	A
6	A	16	C	26	C	36	B	46	A	56	A
7	B	17	D	27	C	37	B	47	C	57	D
8	C	18	B	28	B	38	A	48	D	58	B
9	C	19	C	29	B	39	D	49	C	59	D
10	D	20	C	30	C	40	A	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Both results follow from one idea. A graft simply continues the chosen shoot's own body, so its fruit traits are an unchanged copy of that single parent — a clone — and cannot drift. Seeds, by contrast, form only after gametes fuse in fertilisation, and that fusion reshuffles inherited traits, so seed-grown offspring can vary. The claim that the rootstock passes its fruit traits up is wrong: a graft keeps the scion's own traits, which is exactly why grafting is used. Spores and buds are not involved in either row here. And it is the seed row, not grafting, that shows variation, so the option reversing the two is mistaken.

2. (D) Only sexual reproduction involves two cells (gametes) fusing into one. When the pollen tube carries the male gamete to the egg inside the ovule and they join, that fusion is fertilisation — the one true cell-joining event listed. Budding in yeast is the opposite: a bud grows out and separates, nothing fuses. Fragmentation in Spirogyra means the body breaks into pieces that each grow alone; the pieces do not rejoin. A fern spore is an asexual unit that grows on its own without any fusion, so 'two spores must fuse' is false.

3. (B) The eye of a potato is a bud, and a bud is the part that grows into a new shoot. Piece 2 keeps all its flesh (the food store) but has no bud, so it cannot send up a shoot — proving the bud, not the food, is what was missing. So the conclusion that the eye supplies the food is wrong: the food is in the flesh that piece 2 still has. The eye is not an ovule (ovules form seeds after fertilisation, not vegetative shoots). And piece 2 will not sprout later from flesh alone, because the bud it needs has been removed.

4. (D) A seed forms from an ovule after fertilisation. If the ovary holds no ovules at all, then no matter how perfect the pollination and fertilisation attempt, there is nothing inside that can develop into a seed — which justifies the 'never' prediction. Sticky pollen does not stop seed-set; bees carry sticky pollen well. A long style does not block the pollen tube — tubes routinely grow the full length of the style. And bisexual flowers set seed readily; having both parts is no barrier.

5. (B) Spirogyra's fast multiplication in pond A is fragmentation: the filament breaks into pieces that each grow into a new alga. The thick-walled resistant body in pond B that waits out the dry season and germinates with moisture is a spore. So the two are different modes, not the same. Neither is budding (no outgrowth pinches off here). And neither is a seed — seeds form only after gamete fusion (fertilisation), which is absent in both algae and spores.

6. (A) A seedless banana has no working seeds to sow, so the junior's plan fails at the start. Splitting off shoots from the underground stem is vegetative propagation, which copies the single parent and so yields identical clones — exactly the goal. The specks in the pulp are not spores (and would not become a fungus). Sowing would not give clones even if it worked, because seeds come from fertilisation and can vary. The underground-stem route is asexual, not sexual.

7. (B) With its anthers removed, the flower has no pollen of its own, so any pollen on its stigma came from another plant — that is cross-pollination. For embryos to form, the male gamete must then fuse with the egg: fertilisation. Only after fertilisation do the ovules become seeds. So the minimum chain is cross-pollination, then fertilisation, then seed formation. Self-pollination is impossible with no anthers. Pollination alone never makes seeds — fertilisation must follow. And the order cannot be reversed: fertilisation cannot precede the pollination that delivers the pollen.

8. (C) By Friday, when the stigma is finally receptive, the flower's own pollen is long gone (shed Monday). Bees arriving on Friday can only bring pollen from other flowers, so the seeds form mainly from cross-pollination. The claim that a bisexual flower always self-pollinates first is false here — the timing prevents self-pollen from reaching its own ready stigma. Seeds still form (from other flowers' pollen), so 'unable to form at all' is wrong. And no spores are involved; flowering plants make seeds, not spores.

9. (C) Pollen lands on the stigma; a pollen tube then grows down through the style, enters the ovary, and reaches an ovule where the male gamete fuses with the egg. The anther and filament are the male parts that release pollen, not parts the gamete travels through after landing. The style sits above the ovary, so the route cannot go ovary-then-style. And the filament/stigma order in the last option is jumbled and starts in the wrong part.

10. (D) Each seed develops from one ovule that has been fertilised, and the embryo inside each seed grew from that ovule's zygote. Five seeds therefore mean at least five ovules were present and fertilised. Anthers make pollen, not seeds, so 'five anthers' is wrong. A single ovule does not split into many seeds — each seed needs its own ovule. And zygotes cannot form before pollination, because the male gamete (delivered via pollination) is needed to make a zygote.

11. (B) Cuttings make clones — every plant in field 1 has the same traits, so if the disease beats one, it beats them all together. Seeds form after fertilisation, which reshuffles traits, so field 2 holds varied plants; some happened to carry resistance and survived. The contrast comes from clone-uniformity versus seed-variation, not from spraying (not mentioned), nor from spores (cuttings do not make spores), nor from age (both fields are the same age here).

12. (C) The tassel bears the pollen-making (male) flowers and the cob bears the seed-bearing (female) flowers, so maize has separate unisexual flowers carried on the same plant — not one giant bisexual flower. The pollen comes from the tassel's flowers, not from leaves. And the cob is not just a fruit with no female flower: the silky threads are the stigmas of female flowers that must catch pollen before grains (seeds) can form.

13. (C) The key is the order of events. In layering, a branch develops roots while it is still attached to the parent, and only then is it separated — a form of vegetative propagation. In fragmentation, the body breaks into pieces first and each piece then grows on its own. Since the jasmine rooted before being cut, it is layering, not fragmentation, so the neighbour's 'any cut means fragmentation' rule is false. Buried stems do not make spores, and there is no pollen transfer (so not cross-pollination).

14. (C) Both the moss (spores from capsules) and the Bryophyllum (buds on leaf edges) reproduce asexually, with no fusion of gametes, so they share the same basic kind. The mustard grown from seeds is sexual, since seeds form only after fertilisation. So the matching pair is moss and Bryophyllum. The mustard does not grow from buds, so pairing it with Bryophyllum is wrong. Moss units are spores, not seeds, so pairing moss with mustard as seeds is wrong. And the three are not all alike: the seed route is sexual while the other two are asexual.

15. (D) Pollen is made in the anther (W); the slender stalk holding the anther is the filament (X); the sticky tip that receives pollen is the stigma (Y); and the part housing the ovules is the ovary (Z). The other options misassign the pollen-maker, swap the stalk and the sticky tip, or place the ovary where the anther should be.

16. (C) The bag blocks outside pollen, so the first flower could only use its own pollen — self-pollination — and it set seed. The second flower, with its anthers (its only pollen source) removed and also bagged, had no pollen at all and set no seed. The contrast proves the first flower's own pollen was responsible. The bag rules out a neighbour's pollen, so cross-pollination is excluded. Seeds never form without pollination, and no spores are involved in a flowering plant.

17. (D) After fertilisation an ovule becomes a seed, and the zygote that formed inside it grows into the embryo within that seed. The ovary (not the ovule) becomes the fruit that encloses the seeds. An ovule does not turn into a stigma or into a pollen grain — those are entirely different flower parts with their own origins.

18. (B) Hydra grows a bulge that enlarges and detaches, and yeast grows an outgrowth that pinches off — both are budding, the same process. Moss scatters spores from a capsule (spore formation), and Spirogyra breaks into pieces (fragmentation) — different processes from budding and from each other. So Hydra+yeast is the matching pair. Moss and Spirogyra do not both bud; Hydra does not fragment; and yeast does not release spores from a case.

19. (C) Identical money plants need a clone method, so stem cuttings (vegetative propagation) suit (i). Varied tomato plants for breeding need the reshuffling of fertilisation, so sowing seeds suits (ii). Ferns naturally multiply by spores, so scattering spores suits (iii). The other orders misassign at least one method — e.g. using seeds for the 'identical' batch would give variation, and using spores for tomatoes makes no sense.

20. (C) Bright colour, scent, nectar and sticky pollen are classic adaptations to attract insects, which carry the sticky pollen between flowers. A submerged plant releasing pollen into a current relies on water to carry the pollen — water pollination. So the two use different agents: insects and water. Wind does not suit a nectar-rich showy flower (those features attract animals, not breeze). Both-water is wrong for the showy flower, and insects do not pollinate underwater plants.

21. (D) The base that swells is the ovary, which develops into the fruit — here, the pod. Each pea is a seed, and each seed develops from a fertilised ovule (with its zygote growing into the embryo inside). The stigma, filament and anther are flower parts that do not become the pod. A pea does not grow from an anther or a pollen grain, and a zygote alone is not a seed — it sits within an ovule that becomes the seed.

22. (B) Having both parts only makes self-pollination possible, not certain. A flower whose pollen is shed before its own stigma is ready cannot pollinate itself even though it has both parts — that single real case disproves the 'certain' rule. A flower that always self-pollinates would support the rule, not break it. A flower with no pistil is not even a both-parts flower, so it cannot test the rule. And a spore-maker is not a flower, so it is irrelevant.

23. (A) Fine hairs forming a parachute let the wind carry a light seed far — wind. A thick spongy buoyant husk lets a fruit float and be carried by currents — water. Stiff hooks catch onto fur or clothing so animals carry the fruit away — animals. So the three differ: wind, water, animals. The other orders mismatch at least one feature, for example assigning the floating husk to wind or the hooks to water.

24. (B) Heavy pollen and growing pollen tubes show pollination clearly succeeded, so 'pollination failed' contradicts the evidence. A fruit develops from the ovary only after fertilisation. If no fruit formed despite good pollination, the most reasonable cause is that fertilisation did not happen — the male gamete never fused with an egg. An ovary becoming the fruit is the normal step, not a way to 'skip' it, and seeds form within the fruit, so they cannot be dispersed before any fruit exists.

25. (C) Pollen must first reach the stigma (3, pollination). Then the male gamete fuses with the egg (2, fertilisation). Only after that does the ovary ripen into a fruit holding seeds (1). Finally a seed germinates into a new plant (4). So the order is 3, 2, 1, 4. Fertilisation cannot precede pollination, and the fruit cannot ripen before fertilisation, which rules out the other sequences.

26. (C) Wind delivers pollen blindly in all directions, so only a tiny fraction of grains will ever land on a suitable stigma; the plant must release a vast surplus to make any landings likely. The apple, using insects that fly purposefully flower to flower, needs far less. Wind pollen is light, not heavy. Insects are not the grass's agent, so they are not eating its pollen. And the grass does not court pollinators — wind needs no attracting, which is exactly why grass flowers are plain.

27. (C) It is true neither uses a flower, but that does not make their methods identical. The fern reproduces by spores released from spore cases, while the potato sprouts from buds (eyes) in its tuber — vegetative propagation. These are two different asexual routes. 'No flower means spores' is false, since buds are also flowerless. The fern uses spores and the potato uses buds, so the reversed option is wrong, and the fern is not sexual nor the potato fragmenting.

28. (B) Bagging the unopened flower seals out all outside pollen, so any seeds that form must come from the flower's own pollen on its own stigma — guaranteed self-pollination. Removing the anthers strips away its own pollen and leaving it open invites outside pollen, giving the opposite of what she wants. A breezy, bee-filled spot maximises outside pollen. And scattering spores is meaningless — a tomato makes pollen and seeds, not spores.

29. (B) Sexual reproduction is the one that involves fertilisation, the fusion of gametes, and only the apple grown from a seed fits, since a seed forms after fertilisation. The strawberry runner, the sugarcane cutting, the bread-mould spore and the sweet-potato root sprout are all asexual (no gamete fusion), so the apple card alone is sexual and the claim is correct. A spore is an asexual unit formed with no fusion, so it does not belong in sexual. A runner grows vegetatively from a stem, not from a flower, so it is asexual too. And many plants make seeds, so the apple is not special in that way.

30. (C) Fertilisation produces a zygote as the very first new structure, and that zygote later grows into the embryo housed inside the seed. The seed coat is part of the developing seed, not the first product, and it does not become the fruit. The fruit (from the ovary) forms later and does not become a seed. A pollen grain is not produced by fertilisation at all.

31. (A) Grafting simply continues the shoot's own body on a new root system; the fruit it bears keeps the chosen variety's traits, so no blended new apple appears — the claim is wrong for that reason. Grafting does not fuse gametes (it is vegetative, not sexual). The rootstock does not pass fruit traits up through pollen — that is a misunderstanding of how grafting works. And grafting does not switch the shoot to making spores; it still flowers and fruits as its own variety.

32. (A) Seeds that all land at the parent's base produce a dense cluster of seedlings that must fight the parent and each other for light, water, minerals and space — the central reason dispersal is useful. Closeness to the parent does not stop photosynthesis. Seeds never turn back into spores. And the seedlings here are crowded, not spread out, so 'too much room' is the opposite of the real problem.

33. (B) Sexual reproduction in plants runs through fertilisation, which produces a seed inside a fruit. A plant grown from such a seed therefore arose sexually — that single observation settles it. A bud on a leaf margin, a rooted stem cutting, and a buried underground stem are all vegetative (asexual) routes, so each would show the opposite of sexual reproduction.

34. (D) The stigma and stamens are untouched, so pollen is still made by the anthers and can still land on the stigma — pollination can occur. But seeds develop from ovules, and all the ovules were inside the ovary that was removed, so no seed can form. Pollen is made in the anthers, not the ovary, so removing the ovary does not stop pollen production. The stigma sits atop the style, not on the ovary, so it can still receive pollen. And a stigma cannot turn pollen into seeds by itself — ovules are required.

35. (A) A seed is the product of sexual reproduction: it forms only after gametes fuse in fertilisation. A spore is an asexual unit produced without any gamete fusion. That is the deepest difference. The option swapping them is wrong (the spore is the one with no fusion). A spore does not come from pollination at all. And seeds are not always lighter than spores — spores are typically the lighter, more wind-blown units.

36. (B) Bright colour, scent and nectar attract insects, and such flowers make sticky pollen that clings to visitors — Flower P. Tiny, dull, scentless flowers that dangle and shed clouds of light dry pollen are built for wind, which scatters pollen at random, so a large amount is needed — Flower Q. Wind is not drawn by scent, and bees are not attracted by dull, scentless blooms, so the reversed option fails. Neither flower shows water-pollination features, and Q is clearly not insect-adapted, so 'both by insects' is wrong.

37. (B) The female parts, traced from the receptive top downward, run stigma (where pollen lands), then style (the connecting stalk), then ovary (which holds the ovules that become seeds). Placing the ovary before the style is wrong because the style sits above the ovary. The anther and filament are male parts, not female. And listing ovary first reverses the true top-to-bottom order.

38. (A) Cuttings are clones: each new plant is an exact copy of the one elite parent, and there is no fertilisation to reshuffle traits, so a genuinely new trait will not arise from cuttings alone — the prediction is sound for that reason. Cuttings do not gradually accumulate new traits, and they do not secretly involve a second parent's pollen. The conclusion holds for any crop grown only from cuttings, not just a seedless one.

39. (D) Self-pollination needs no outside pollinator or second plant, so it lets a plant set seed reliably even when pollinators are scarce. Cross-pollination mixes pollen from another plant, combining two parents' traits and so producing more varied offspring. The option that swaps these benefits is wrong: it is cross-pollination, not self, that gives variety, and self, not cross, that guarantees seed set without a pollinator. Neither habit makes spores in a flowering plant. And the two habits do not give identical offspring, since cross-pollination adds variation.

40. (A) Fertilisation is the fusion of the male gamete with the egg, and that male gamete arrives inside pollen. So the pollen must first be delivered to the stigma (pollination) before any gamete can reach and fuse with the egg — pollination necessarily precedes fertilisation. The ovary ripening into a fruit happens after fertilisation, not before. Petals do not need to fall before pollen lands. And seed dispersal is the last step of all, long after fertilisation.

41. (B) Both fruits develop the same way: an ovary becomes the fruit and each fertilised ovule becomes a seed. The only difference is how many ovules the ovary held — many in one flower, just one in the other. So no different process is involved. A fruit is itself the product of sexual reproduction (a flower being fertilised), so neither one-seeded fruit comes from vegetative propagation, a spore, or an asexual route.

42. (A) Hooked fruits clinging to socks are carried by animals (or people). Sweet berries eaten by a bird whose gut later passes the hard seeds far away are also animal-dispersed. A dried pod that snaps and flings its own seeds is explosion (self-bursting), needing no outside carrier. So the order is animals, animals, explosion. The berries are not wind-borne, and the bursting pod is not wind or water dispersal, so the other options mismatch at least one case.

43. (D) Self-pollination needs the flower's own pollen, but the anthers (its only pollen source) were removed, so it had no pollen to use; and the bag kept all outside pollen out. With no pollen at all, no fertilisation and so no true seed-bearing fruit could form — so calling it self-pollination is impossible, and a genuine seed-set here would mean the observation was mistaken. The bag could not trap pollen that was never there, it blocks (not admits) neighbour pollen, and darkness does not trigger spores in a flowering plant.

44. (D) A potato multiplies by buds (eyes) in its tuber — vegetative propagation — not by spores, so that pairing is the wrong one. Rhizopus (bread mould) genuinely reproduces by spores, Spirogyra genuinely by fragmentation, and yeast genuinely by budding, so those three pairs are all correct.

45. (A) A unisexual male flower has only stamens — no pistil, ovary or ovules — so it has nothing to be fertilised and cannot form seeds on its own. It certainly can release pollen, since it has anthers (that is what makes it male). Its pollen can be carried to a female flower, so it does take part in pollination. And it is not sterile: its pollen fertilises female flowers, so it contributes to reproduction, just not by forming seeds itself.

46. (A) Roots (dahlia), bulbs which are shortened underground stems (onion), and leaf-and-bud cuttings (rose) are all vegetative routes, growing a new plant from a part of the parent body. A seed is different: it forms only after fertilisation and is a product of sexual reproduction, so the sunflower's seed is the non-vegetative route. The other claims are false, since a dahlia root does give a new plant, an onion bulb is an underground stem (not a fruit), and a rose cutting grows from the stem itself, not from a hidden seed.

47. (C) After fertilisation, each ovule becomes a seed, the ovary becomes the fruit that encloses the seeds, and the zygote (inside the ovule) grows into the embryo. The other options swap these fates — for example making the ovary into a seed, the ovule into an embryo, or the zygote into a fruit — all of which are incorrect.

48. (D) Cuttings from one vine are all clones, so they carry the same traits; the differences in size and colour are caused by their surroundings (light, soil), not by any change in their inherited traits. So the claim that they are now genetically different is wrong. The environment does not permanently rewrite a plant's traits, poor soil does not switch a plant to sexual reproduction, and the plants grew from cuttings of one parent, not from varied seeds.

49. (C) A submerged plant releasing pollen into a current relies on water to carry it — water pollination. A grass with loose, dangling anthers shaking pollen into the air relies on wind — wind pollination. So the two differ: water and wind. Neither plant shows insect-attracting features (no colour, scent or nectar), and the grass on the bank shaking pollen into the breeze is wind-borne, not water-borne, so the other options mismatch.

50. (B) All ten flowers were pollinated, but only those whose pollen tubes could reach the ovules (allowing fertilisation) set fruit; the half where fertilisation was blocked set none. This shows pollination by itself is not enough, fertilisation must follow for a fruit to form. The failure was due to blocked fertilisation, not old pollen. The blocked flowers did not form seedless fruit, they formed no fruit at all. And no spores are involved, since a flowering plant makes seeds through fertilisation.

51. (A) Stem cuttings are a quick vegetative method that copies the single parent exactly, giving identical clones in one season — meeting both aims. A seedless plant has no usable seeds to sow, and even if it did, seeds vary, failing the 'identical' aim. Cross-pollination by wind would mix in another plant's traits, again failing 'identical.' And the ornamental does not make spores, so waiting for spores is futile.

52. (D) The new plant begins at fertilisation, when the male gamete fuses with the egg, which the diary records on Day 3; that fusion produces the zygote noted on Day 4. Day 1 is only pollination (pollen reaching the stigma), which comes before the new plant starts. Day 20 (ripe fruit) is far too late, long after the embryo has begun. And the new individual dates from the fusion on Day 3, not merely from when the resulting zygote becomes visible on Day 4.

53. (D) Wind carries seeds best when they are light and offer surfaces to catch the air. A heavy seed with a thick stony coat is dense and offers no sail, so wind can barely move it — the least helpful feature. Fine parachute hairs, flat papery wings, and a small light body all help a seed stay aloft and travel on the wind.

54. (A) Seeds develop from ovules. With no ovules at all, the flower has nothing that can become a seed even after a pollen tube reaches the ovary — so it cannot form seeds. Missing ovules do not block the anthers, which still release pollen. The stigma exists independently of ovules, so it can still receive pollen. And insects respond to colour, scent and nectar, not to whether the hidden ovary holds ovules, so they will still visit.

55. (A) During pollination, pollen lands on the stigma, the sticky receptive tip of the pistil. The filament is merely the stalk that holds the anther up, so it is not where pollen lands. The filament is not sticky, so the statement is not correct. Pollen is produced and stored in the anther but lands on the stigma — landing 'on the anther' confuses production with reception. And pollen does not land inside the ovary; it must travel there via a pollen tube only after landing on the stigma.

56. (A) Carrying pollen to the stigma of a different plant of the same kind is cross-pollination, and deliberately crossing two chosen varieties to combine their traits is hybridisation. The transfer step is not fertilisation (that is the later fusion of gametes), and it is not self-pollination (the pollen comes from another variety). Fragmentation and budding are asexual modes, unrelated to crossing two varieties.

57. (D) Vegetative propagation means a new plant grows from a part of the parent body (here, a bud on an underground stem) with no seed, fruit or fertilisation involved — that is the convincing sign. A unit formed inside a fruit after fertilisation is a seed, which signals sexual reproduction, the opposite. Bees visiting flowers points to pollination and seeds, again sexual. And an embryo in a seed coat is exactly what a seed contains, so that too indicates sexual reproduction.

58. (B) A juicy, colourful, sweet fruit is an invitation to animals: they eat it and later drop or pass the hard, undigested seeds far from the parent — animal dispersal. Bright colour does not make a fruit light enough to fly, and such juicy fruits are not built for wind. The juice does not turn the fruit into a floating, water-dispersed body (and it is far from water). Sweetness does not cause bursting; explosive dispersal comes from drying pods, not sweetness.

59. (D) In budding, a single small outgrowth (bud) develops on the parent and then pinches off as a new individual — the parent body stays whole. In fragmentation, the parent body itself breaks into several pieces, each of which grows into a new individual. The mere fact that 'something separates' does not make them identical. The reversed option mislabels which process does what. And neither is spore formation, which would mean spores released from a case, not seen here.

60. (A) Seeds form after fertilisation, where gametes fuse and traits are reshuffled, so seed-grown plants can show variation, including occasionally a useful new trait. Cuttings clone the single parent with no such reshuffling, so they show none. Cuttings do not make spores or erase traits. Seeds do not skip fertilisation — fertilisation is exactly what makes them. And it is seeds (sexual) that involve two parents' gametes, not cuttings; the option has it backwards.